Report CA1

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```
sort Data;
cons d1, d2: Data;

act  send, save, remove, read : Data;
     comm_put: Data;
     comm_get: Data;

proc Producer = send(d1).Producer + send(d2).Producer;

proc Consumer = sum d: Data. read(d) . Consumer;



proc Buffer0(N: Nat) = sum d: Data. save(d) . Buffer1(d, N);

%proc Buffer1(x: Data, N: Nat) =
%  if(N > 1,
%     (sum d: Data .save(d).Buffer2(x, d, N)) + (remove(x).Buffer0(N)),
%     remove(x) . Buffer0(N)
%  );

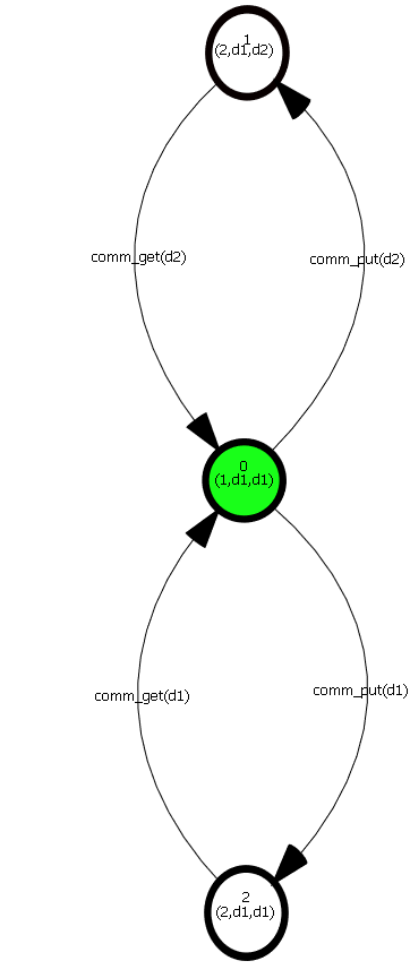
proc Buffer1(x: Data, N: Nat) =
       (N > 1) -> (
         (sum d: Data. save(d) . Buffer2(x, d, N)) + (remove(x).Buffer0(N))) <>
         (remove(x) . Buffer0(N));

proc Buffer2(x: Data, y: Data, N: Nat) = remove(x).Buffer1(y, N);


init allow({comm_put, comm_get},
       comm(
         {send | save -> comm_put,
          remove | read -> comm_get
         },
         Producer || Buffer0(2)  %toggle the size to 1 or 2
         || Consumer
       )
     );
```

we specify the size of buffer here

خروجی کد N=1 :



خروجی کد N=2

